Photopheresis in Adults and Pediatrics

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66.1 Introduction

Extracorporeal photopheresis (ECP) is a leukapheresis-based treatment that has been used during the last decades by many clinicians. Based on the results of a prospective, multicenter, international clinical trial in patients with cutaneous T-cell lymphoma (CTCL), ECP was approved by the FDA as the first cellular immunotherapy for cancer in 1988 (Edelson et al. 1987). During the last decades, ECP has been investigated worldwide for the prevention and treatment of a variety of T-cell-mediated diseases including acute and chronic GVHD, solid organ and tissue transplantation, systemic sclerosis, systemic lupus erythematodes, and Crohn's disease (Knobler et al. 2020, 2021). Administering ECP to patients suffering from these diseases revealed promising results both in prospective and retrospective single and multicenter clinical studies. Despite its frequent use, the mode of action of ECP remains elusive but includes reduction of pro-inflammatory cytokines, induction of anti-inflammatory cytokines, and modulation of immune cell populations.

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66.2 Technical Aspects

During ECP, the patient's blood is collected via an antecubital vein or a permanent catheter, and the white blood cells are separated from the red blood cells and plasma by centrifugation in a device that is specifically constructed for the procedure (Knobler et al. 2020, 2021; Schoonemann 2003). Collected mononuclear cells (MNCs) using either continuous or discontinuous cell separators are then exposed ex vivo to a photosensitizing agent, 8-methoxypsoralen (8-MOP), which is added directly to the buffy coat/plasma fraction followed by photoactivation with ultraviolet A (UV-A) irradiation and then reinfusion of the photoactivated product (Schoonemann 2003).

ECP has originally been developed as a single procedure that combines the separation of the MNCs from the whole blood with irradiation of the 8-MOP-treated leukapheresis products within a single machine ("closed system of ECP"). The "offline technique" (two-step method) of ECP treatment includes as the first step cell separation with a standard blood cell separator that can also be used for the collection of peripheral blood stem cells. The apheresis product is transferred into another disposable, 8-MOP is added, and irradiation is performed with a separate machine at a dosage of 2 J/cm^2. After irradiation, transfusion of the treated cells is carried out manually by a standard transfusion set. Both ECP techniques have demonstrated clinical efficacy, but almost all clinical studies have been performed with the

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single ECP technique, and studies comparing both systems are almost completely lacking (Schoonemann 2003; Andreu et al. 1994; Brosig et al. 2016).

66.3 Results of ECP in Acute GVHD

Until recently, no consensus on the optimal choice of agents for salvage therapy of steroid-refractory acute GVHD has been reached, and treatment choices are based on the physician's experience, risk of toxicity and potential exacerbation of pre-existing comorbidity, interactions with other agents, and ease of use (Martin et al. 2012). During the last years, more and more HCT centers have administered ECP to patients with steroid refractory acute GVHD. Results of larger prospective studies on the use of ECP in this indication are shown in Table 66.1. The intensified schedule of ECP with two to three treatments per week on a weekly basis significantly improved response rates in patients with GI involvement and grade IV acute GVHD (Greinix et al. 2006).

In a systematic review of prospective studies including 6 studies with 103 patients given ECP for steroid-refractory acute GVHD, an overall response rate (ORR) of 69% was achieved including ORR for skin, liver, and GI involvement of 84%, 55%, and 65%, respectively (Abu-Dalle et al. 2014). Compared to anticytokine treatment, administration of ECP for steroid-refractory acute GVHD not only achieved significantly higher ORR (66% vs. 32%) and CR (54% vs. 20%), but ECP was also an independent predictor of response and survival and was associated with significantly lower NRM and superior survival in steroid-refractory grade II acute GVHD (Jagasia et al. 2013). Compared to other IST, ECP has an excellent safety profile with limited toxicity concerns, no increased concerns for viral reactivations, and no documented interaction with other drugs (Martin et al. 2012).

In May 2022, the European Medical Agency (EMA) approved ruxolitinib for the treatment of patients aged 12 years and older with acute GVHD who have inadequate response to corticosteroids or other systemic therapies based on the results of the REACH2 study comparing ruxolitinib with best available therapy in patients with steroid-refractory acute GVHD (Zeiser et al. 2020). Therefore, ruxolitinib has been frequently used as second-line therapy during the last year. In a recently published single-center experience of combining ruxolitinib with ECP in 18 patients

Table 66.1 Results of second-line treatment of acute GVHD using extracorporeal photopheresis

Author (year)	No. of patients	CR skin no. (%)	CR liver no. (%)	CR gut no. (%)	OS%
Salvaneschi et al. (2001)	9	6/9 (67)	1/3 (33)	3/5 (60)	67
Dall'Amico (2002)	14	10/14 (71)	4/7 (57)	6/10 (60)	57
Messina et al. (2003)	33	25/33 (76)	9/15 (60)	15/20 (75)	69 at 5 y
Greinix et al. (2006)	59	47/57 (82)	14/23 (61)	9/15 (60)	47 at 5 y
Garban et al. (2005)	12	8/12 (67)	0/2 (0)	2/5 (40)	4
Kanold et al. (2007)	12	9/10 (90)	5/9 (56)	5/6 (83)	75 at 8.5 m
Calore et al. (2008)	15	12/13 (92)		14/14 (100)	85 at 5 y
Perfetti et al. (2008)	23	15/23 (65)	3/11 (27)	8/20 (40)	48 at 37 m
Gonzalez-Vicent et al. (2008)	8	8/8 (100)	2/2 (100)	4/7 (57)	38
Perotti et al. (2010)	50	39/47 (83) (1)	16/24 (67) (1)	8/11 (73) (1)	64 at 1 y
Jagasia et al. (2013)	57	38/57 (67) (1)	38/57 (67) (1)	38/57 (67) (1)	59 at 2 y
Calore et al. (2015)	72	50/64 (78)	10/12 (84)	42/55 (76)	71 at 5 y
Niittyvuopio et al. (2018)	52	24/31 (77)	1/3 (33)	14/41 (34)	51 at 1 y
Worel et al. (2018)	99	71/88 (80)	18/29 (61)	26/35 (75)	50 at 5 y
Amat et al. (2021)	37	9/28 (32)	3/11 (27)	12/30 (40)	80 at 3 y

Abbreviations: *No* number, *CR* complete resolution, *OS* overall survival, *y* years, *m* months
Results were provided as complete and partial resolution

with severe SR acute GVHD of lower GI tract (50% each overall grades III and IV) and other organ manifestations (skin $n = 7$, liver $n = 6$, and upper GI tract $n = 2$), the majority of patients ($n = 15$, 83%) received ruxolitinib a median of 20 days before starting ECP (Modemann et al. 2020). The investigators observed a best ORR of 55% including a CR rate of 44% and an additional PR rate of 11%, respectively. The mean daily steroid dose was 130 mg at the diagnosis of SR acute GVHD and at the start of lead-in ruxolitinib (83% of patients) or ruxolitinib with ECP (17% of patients) treatment and could be tapered to less than 20 mg by day 21 and stopped after a median of 27 days. Thus, the feasibility of a rapid steroid taper and discontinuation of steroids with the combination of ruxolitinib and ECP could be demonstrated. Responding patients had a two-year OS of 70% with a median survival of 18 months. These promising results should be confirmed in prospective studies to increase the treatment options for patients with severe SR acute GVHD.

66.4 Results of ECP in Chronic GVHD

Many therapeutic options have been reported for salvage treatment of steroid-refractory chronic GVHD, and until recently, no single class of IS agent has been established as standard therapy (Wolff et al. 2011). In August 2017 ibrutinib and in September 2021 ruxolitinib were approved by the FDA for the treatment of patients with chronic GVHD after failure of one or more lines of systemic therapies.

ECP represents a frequently used therapeutic approach for the treatment of chronic GVHD patients failing corticosteroids (Table 66.2) (Knobler et al. 2020, 2021; Wolff et al. 2011; Greinix et al. 1998; Flowers et al. 2008; Jagasia et al. 2009; Greinix et al. 2011). Most of the clinical experience in ECP treatment of steroid-refractory chronic GVHD patients is based on retrospective analyses with consistently high response rates in up to 80% of patients with cutaneous manifestations and substantial improve-

Table 66.2 Results of the use of extracorporeal photopheresis in chronic GVHD

Author (year)	No of patients	CR/PR skin (%)	CR/PR liver (%)	CR/PR oral (%)	ORR (%)
Greinix et al. (1998)	15	80	70	100	Na
Salvaneschi et al. (2001)	14	83	67	67	64
Messina et al. (2003)	44	56	60	–	57
Seaton et al. (2003)	28	48	32	21	36
Apisarnthanarax et al. (2003)	32	59	0	Na	56
Foss et al. (2005)	25	64	0	46	64
Rubegni et al. (2005)	32	81	77	92	69
Greinix et al. (2006)	47	93	84	95	83
Couriel et al. (2006)	71	57	71	78	61
Kanold et al. (2007)	15	75	82	86	50
Perseghin et al. (2007)	25	67	67	78	73
Flowers et al. (2008)	48	40	29	53	40
Jagasia et al. (2009)	43				65
Perotti et al. (2010)	23	96	100	80	69
Dignan et al. (2012)	82	92	Na	91	74
Greinix et al. (2011)	29	31	50	70	Na
Del Fante et al. (2016)	102	Na	Na	Na	81
Ussowicz et al. (2013)	13	67	89	86	69
Hautmann et al. (2013)	32	59	100	60	44
Dignan et al. (2014)	38	65	–	29	50
Sakellari et al. (2018)	88	83	53[a]	Na	73
Gandelman et al. (2018)	77	55	Na	Na	62

Abbreviations: *No* number, *CR* complete resolution, *PR* partial resolution, *ORR* overall response rate, *na* not available
[a]Visceral involvement

ment in sclerodermatous skin involvement (Knobler et al. 2020, 2021; Wolff et al. 2011).

In a multicenter, randomized, controlled, prospective phase II study of ECP in 95 patients with steroid-refractory/dependent/intolerant chronic GVHD, significantly more patients in the ECP arm achieved a complete or partial response of cutaneous manifestations ($p < 0.001$) as well as a 50% reduction in steroid dose and at least a 25% decrease in total skin score ($p = 0.04$) by week 12 (Greinix et al. 1998). A steroid-sparing effect of ECP has also been reported by other investigators (Knobler et al. 2020, 2021; Wolff et al. 2011; Flowers et al. 2008; Jagasia et al. 2009).

In a systematic review of prospective studies on the use of ECP in patients with chronic GVHD, an ORR of 71% in cutaneous, 62% in GI, 58% in hepatic, 63% in oral mucosal, and 45% in musculoskeletal manifestations of chronic GVHD was reported (Abu-Dalle et al. 2014). The rate of IS discontinuation was 23%, and ECP was tolerated excellently. In another meta-analysis, high response rates in cutaneous and extracutaneous manifestations of chronic GVHD including 48% of responses in lung involvement were confirmed (Del Fante et al. 2016). The ECP schedule in chronic GVHD is empirical ranging from multiple treatments per week on a weekly basis to two treatments biweekly and in case of response prolongation of the treatment interval to 4–6 weeks, respectively. No clear association between ECP dose intensity and response has been reported. Higher response rates were achieved in steroid-refractory patients given ECP earlier in the course of their disease (Malik et al. 2014; Messina et al. 2003). Improvements in the quality of life and survival in ECP responders have been reported (Knobler et al. 2020, 2021; Wolff et al. 2011; Greinix et al. 1998; Malik et al. 2014; Messina et al. 2003).

In a retrospective analysis, 23 patients with SR chronic GVHD (57% NIH grade 3, 91% beyond second-line treatment, and 87% with more than one organ involved) received the combination of ruxolitinib at 5–10 mg bid and ECP with two treatments on consecutive days every 2–4 weeks (Maas-Bauer et al. 2021). Thirty-five percent of patients started ECP and ruxolitinib treatment simultaneously, whereas 30% started ECP first and the median time of ECP therapy prior to combination treatment was 3.25 (1–7) months. During ECP alone, the best response was PR in 43% (3/7) of patients and 57% (4/7) were nonresponders. Thirty-five percent of patients started ruxolitinib treatment first a median of 15 (1–29) months prior to combination treatment. The best ORR of ruxolitinib alone was PR in 62.5% (5/8), and 37.5% (3/8) did not respond. The best ORR of ECP combined with ruxolitinib was 74% (17/23) including 9% CR and 65% PR and a two-year OS of 75%. Thus, combinational treatment increased ORR in heavily pretreated patients with multiorgan involvement SR chronic GVHD and was able to improve the outcome of patients after inadequate responses to ECP or ruxolitinib monotherapy. Patients received a median of six months of combination therapy.

ECP is a safe and efficacious treatment for patients with chronic GVHD with steroid-sparing capacity. Transient hypotension during treatment and mild anemia and/or thrombocytopenia have been reported as side effects of ECP. Prospective clinical studies are warranted to assess the efficacy of ECP in well-defined cohorts of chronic GVHD patients treated earlier in the course of their disease. Recently, Jagasia and colleagues reported the first results of a randomized, controlled, multicenter study in NIH-defined moderate/severe chronic GVHD patients given ECP in the study arm in combination with standard of care IS (Jagasia et al. 2017). Besides an ORR of 74%, and thus, a promising efficacy ECP demonstrated to be safe and tolerated well.

66.5 Conclusions

ECP has been used for over 30 years in the treatment of CTCL, acute and chronic GVHD, and solid organ transplant rejection. Multiple scientific organizations recommend its use due to ECP's efficacy and excellent safety profile

(Knobler et al. 2020, 2021). Due to the lack of interactions with other agents and the avoidance of general IS, ECP compares favorably with other IS strategies, supporting its increasingly frequent use as salvage therapy of steroid-refractory/dependent acute and chronic GVHD. Of note, the corticosteroid-sparing potential of ECP has been confirmed in numerous retrospective and prospective studies and translates into immediate clinical benefit for patients with GVHD as well as a reduction of transplant-associated morbidity and mortality.

No general recommendation can be made on the treatment schedule due to missing evidence. Ideally, ECP treatment should be initiated as early as possible after the indication is confirmed. Especially in patients with steroid-refractory acute GVHD, earlier treatment onset and an intensified weekly ECP schedule resulted in improved response rates and patients' outcomes. Prospective studies on the use of ECP as upfront treatment in GVHD as well as in combination with recently approved agents are warranted.

Key Points

- ECP is a safe and efficacious adjunct therapy of steroid-refractory acute and chronic GvHD.
- Results in upfront therapy of chronic GvHD are promising.

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